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RESEARCH ARTICLE

A Review of Motors With Reconfigurable Windings for Automotive Traction Drives

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ABSTRACT This paper provides an overview of the application of reconfigurable windings in traction motors and their prospective benefits for electric vehicles. Methods of implementation discussed are series–parallel, star-delta, and tapped windings with series-parallel configurations having the greatest potential for system performance improvements. The review provides valuable insights into the impact on e-drive system mass and volume, showing significant improvements to torque and power densities of approximately 40-80% can be achieved. Other investigations targeting motor efficiency show total motor loss reductions of approximately 30-60% are also possible. The published literature on the topic tends to be case-specific, indicating that the generalisation of the technology applicability is lacking, with a particular need for design tools and methodologies early in the propulsion system design stage, to explore how the technology could be best applied. This is also highlighted by the system level impacts shown in a case study on the Nissan Leaf drive, with series-parallel reconfigurable windings resulting in a 41.67% inverter current reduction while maintaining baseline performance characteristics. Other key challenges to the technology uptake include the cost and complexity of the reconfiguration device, however If these can be overcome, and the design advantages better understood, traction drives with reconfigurable windings show potential in reducing the size, weight, and cost of the electric machines, while improving their efficiency and power density.

INDEX TERMS Automotive propulsion, efficiency and power, electric machines, extended speed range, PMSM, torque density, winding reconfiguration.

I. INTRODUCTION

In recent years, the automotive industry has witnessed a significant shift towards electrification leading to a growing demand for advanced electrical systems and components. Electric traction drives, play a crucial role in the overall performance and efficiency of the electric vehicle (EV), the design of which is constantly evolving to meet the increasing demands of the automotive sector [1], [2], [3], [4], [5]. Incoming regulations governing resource depletion and life cycle emissions place a greater emphasis on the use and acquisition of materials specific to the electric powertrain, translating to a heightened emphasis on downsizing the battery, motor-generator, and inverter [6], [7].

This paper aims to provide an overview of the applications of reconfigurable windings in the automotive sector. The review will explore the concept of reconfigurable windings, their advantages, limitations, and two main approaches of implementing them in electric machines: 1. Reducing drive system mass and volume, thus improving torque and power densities, and 2. Reducing total drive losses under operation, contributing to an improved vehicle efficiency. Further considerations and challenges of the implementation of this technology are also provided. A case study of the Nissan Leaf drive demonstrates the application of approaches 1 & 2, to significantly reduce inverter current demand while maintaining baseline performance characteristics. Although this

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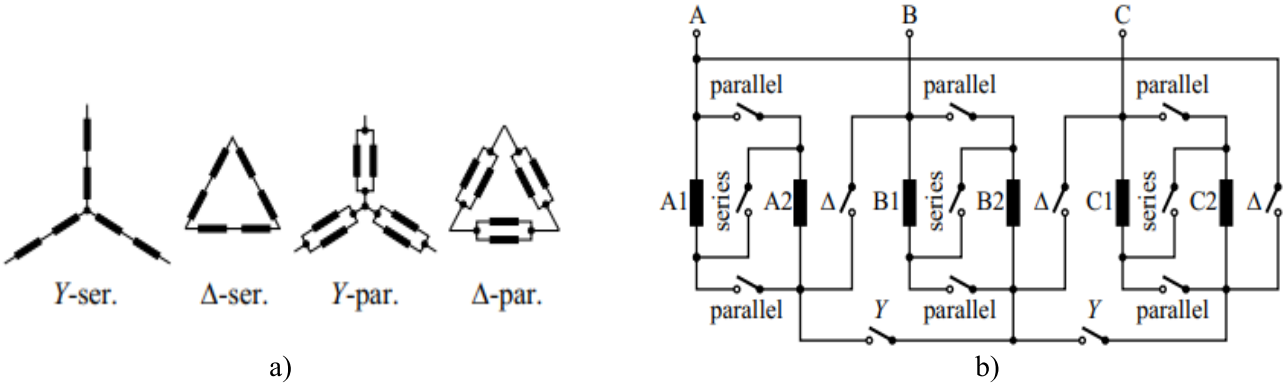


FIGURE 1. a) Possible winding configurations available with two winding parts per phase. b) Suggested switching diagram required to achieve the series-parallel and star-delta reconfiguration methods for a three-phase machine [13].

paper focuses on automotive applications, differing motor types and drives may find relevance in these conclusions with previous examples for marine applications [8], industrial equipment [9], [10], [11], and even domestic washing machines [12].

II. RECONFIGURABLE DRIVE CONCEPTS

The term reconfigurable windings refers to a capacity to alter the armature winding connections of an electric machine. This acts to effectively change the number of turns per phase during operation. Primarily, this provides another dimension of flexibility and adaptability to the design as each additional winding configuration, demonstrated in Figure 1, expands the operating conditions for the machine [12]. The means of reconfiguration can be achieved by subdividing the windings of each phase into two parts and connecting them in either Series or Parallel. Altering the termination of the phases between Star and Delta also influences the machine characteristics in the same manner, albeit to a lesser extent. By combining both the Series-Parallel and Star-Delta reconfiguration methods, as in Figure 1, four 'balanced' configurations are possible from two winding parts per phase [13].

To achieve multiple configurations during operation (with a reasonable switching transient) additional reconfiguration equipment is required, often depicted using semiconductors [10] or occasionally mechanical contractors [14] as the switching components. An early example of the required circuit (Figure 1b) capable of 4 different winding configurations totals 14 switching components. Because of the complexity, and switch size necessary to carry the full phase current, such a device is often represented as separate to the machine and similar in size to the drive electronics. Furthermore, a greater number of stator leads terminating outside of the machine are also required; when considering a conventional three-phase machine originally with 3 stator leads, the capability to switch between Star-Delta configurations increases the required stator leads to 6, or to 12 for Series-Parallel reconfiguration, all of which are represented later in Table 1.

For automotive traction machines, torque density is one of the most important design parameters and is achieved by maximising the magnetic forces between the rotor and stator for the available design volume and or mass. The first step in maximising the torque of an electric machine is to produce the highest possible flux density. Beyond constructing a 'good' magnetic circuit and choosing high-quality magnetic materials, as many conductors as possible should be packed into the available space providing a high slot fill factor. Secondly, each conductor must carry as much current as possible without reaching the thermal limit of themselves or surrounding materials [15].

A. ELECTROMAGNETIC TORQUE

The force generated by the conductors in the magnetic circuit is not directly bound by the manner in which they are added to the slot, i.e. a high number of turns (N) of a thin conductor carrying a low current, (I) will produce the same reaction force as a low number of turns of thick conductors carrying a high current. This product of turns and current can be expressed in terms of magnetomotive force mmf:

$$mmf = NI, \phi = \frac{mmf}{R} = \frac{NI\mu}{g} \tag{1.1}$$

where ϕ is flux, R is reluctance, N is the number of turns, I is current, μ is the magnetic permeability of air, and g the airgap length.

Providing that the product of NI remains constant, the level of flux ϕ and thus ability for differing winding configurations to produce torque will be the same. It is therefore useful when comparing a machines torque capability, to refer to mmf in the literal sense; as ampere-turns (At).

$$T = \left(\sum BIL\right) r, \quad B = \frac{\phi}{A} \tag{1.2}$$

where T is torque, B is the flux density, A is the cross-sectional area of the flux path, I is the armature current, L is the slot length, and r is the acting radius of the rotor.

Following the generalised equation for electromagnetic torque (1.2) with a fixed armature current, increasing the

number of turns (N) proportionally increases the electromagnetic torque of the machine. Therefore, it should be expected that when changing from a parallel to series configuration, as the series connection features double the number of turns, double the flux and electromagnetic torque can be produced for the same armature current. However, armature winding current densities and magnetic saturation of the iron pose a realistic limitation to the torque density of the machine.

B. RESISTANCE AND INDUCTANCE

When changing the number of turns in an electric motor, both phase resistance and inductance are affected proportionally to the square of the number of turns. Therefore, if the number of turns is halved, as when reconfiguring from series to parallel connection, the inductance and resistance of the machine will decrease by a factor of 4.

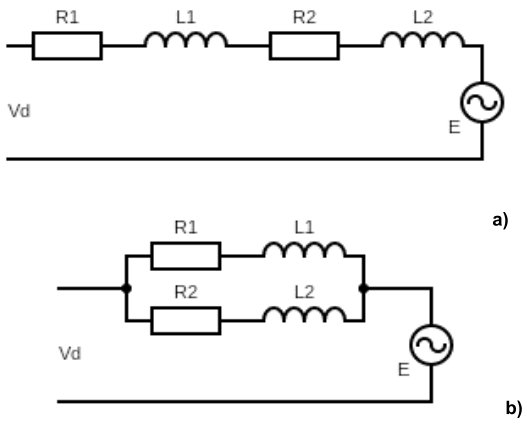


FIGURE 2. Equivalent circuit diagram for a) series and b) parallel winding configurations.

This relationship can be demonstrated when looking at the equivalent circuit shown in Figure 2. For the series connection, the inductances and resistances are the sum of both parts whereas the parallel connection has an inverse relationship.

	Unbalanced Windings	Balanced Windings
Series connection resistance and inductance:	$R_s = R1 + R2$ $L_s = L1 + L2$	$R_s = 2R$ $L_s = 2L$
Parallel connection resistance and inductance:	$\frac{1}{R_P} = \frac{1}{R1} + \frac{1}{R2}$ $\frac{1}{L_P} = \frac{1}{L1} + \frac{1}{L2}$	$R_P = \frac{1}{2}R$ $L_P = \frac{1}{2}L$

(1.4)

If each winding part is balanced and of the same resistance and inductance (as is advantageous to avoid circulating currents) this is simplified further to a ratio of 4 between series and parallel connections. A lower winding inductance and resistance has a tendency to increase motor efficiency, in part due to the influences on copper losses, iron losses, and reactive power [13]. This partially provides insight as to why an increase in torque at higher rotational speeds may be available when changing from series to parallel configurations as shown in Figure 3. This influence is highly sensitive to specific machine and drive applications. Nevertheless, a reduction in machine inductance and resistance is also reflected in technology roadmaps, anticipating high speed electric machines [16]. Equally, there are disadvantages with excessively low inductances, namely increased harmonics necessitating higher switching frequencies and further motor control and stability considerations [13].

C. SATURATION

A realistic bound to the total flux in a machine is commonly known as the saturation limit. Magnetic saturation can be defined by an increase in ***mmf*** not producing a proportional increase in the total flux. Applying this to equation **(1.1)**, it is logical that for this to be true, an increase in reluctance must be present. This can be attributed to the properties of the magnetic steel forming the stator and rotor. Although usually having negligible reluctance, it rapidly increases beyond a certain flux density (commonly 1.8-2.0T) indicating that machine is 'saturated'. It is generally understood that operation in magnetic saturation, although possible, should be avoided due to the increased losses and thermal load on the machine, along with a decreasing return in output torque and efficiency [15]. This can offer explanation as to why an increased number of turns may not in fact produce a proportional increase in torque. This forms a practical limit to the desired ***mmf*** of the machine without incurring a higher iron mass.

D. BACK-ELECTROMOTIVE FORCE

It is previously stated that the electromagnetic torque capability characterised by **(1.2)**, where the product of the number of turns (N) and armature current (I) is proportional to torque. This is only true however up to the base speed of the machine, depicted in Figure 3, due to the back-electromotive force (e) generated in the phase approaching the same value as the DC bus voltage. At this point, as there is no voltage difference across the motor to the bus voltage, the controller can no longer drive the machine with q-axis current to the motor and drive it any faster. This back EMF can be estimated from Faraday's Law:

$$e_{ph} = -N\frac{d\phi}{dt} \tag{1.3}$$

where, N is the number of turns and ϕ is the airgap flux.

The back emf produced by each phase of an electric machine has the characteristic of being proportional to the angular velocity of the rotor, the number of turns (N) present in the armature windings, and the airgap flux density (ϕ). From **(1.1)** as the airgap flux is also a function of the **m.m.f**, an increase in torque from a higher number of turns also

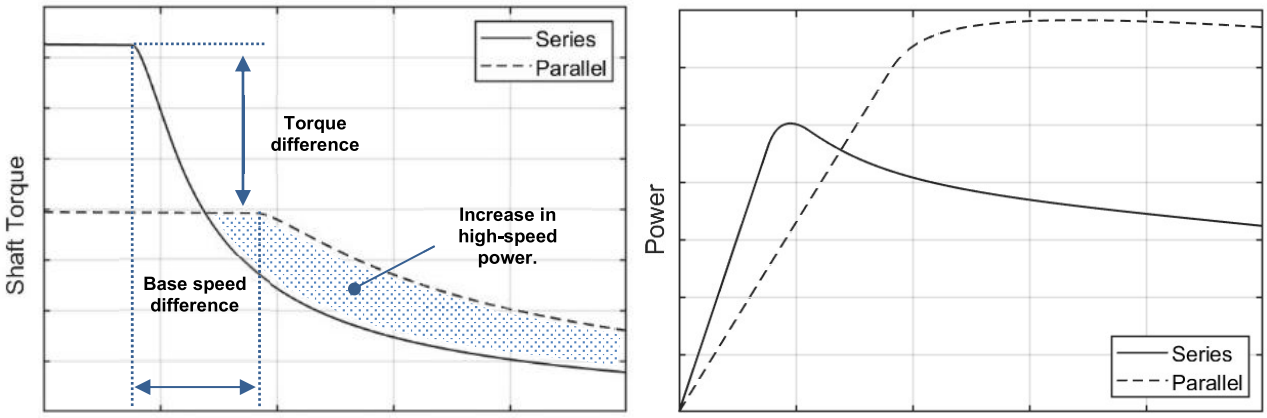


FIGURE 3. The generalised torque profile for a machine with reconfigurable windings. Each phase is subdivided into 2 parts, the series connection providing a higher torque capability, parallel a higher power output.

proportionally increases the back emf of the machine. Using **(1.3)**, increasing the number of turns with a fixed armature current will increase the magnitude of constant torque, but also decease the base speed such that the power output of the drive is maintained. Hence, this electromagnetic relationship of trading torque and speed can be likened to that of a mechanical gearbox.

E. SALIENCY AND FIELD WEAKENING

Saliency refers to the asymmetry or variation in reluctance of the rotor. It occurs when the rotor (the rotating part of the motor) is deliberately designed to have a non-uniform magnetic field distribution. This non-uniformity creates a difference in the magnetic reluctance along different paths within the motor as shown in Figure 4.

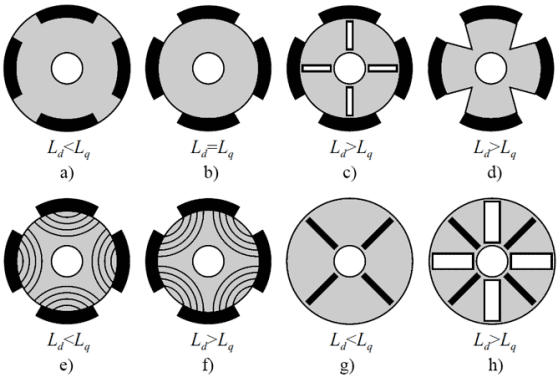


FIGURE 4. Differing rotor structures and saliency ratios where black represents magnet material and grey is the rotor iron [13].

This reluctance is generally measured as the resultant inductance in the stator windings in two coordinates - the direct (d) axis, normal to the flux of the rotor's magnetic field, and the quadrature (q) axis, orthogonal, or 90° electrical degrees from the direct axis, which generates the torque [13]. As the magnet material has a similar reluctance to air, a surface mounted permanent magnet (SMPM) rotor will have negligible variance in reluctance, and because of this, is considered 'non-salient.'

Rotor saliency primarily influences the speed range capability of an electric machine and torque produced by either the magnets or reluctance of the rotor structure [17]. By precisely knowing the rotor position, a negative d-axis current can be applied to the stator windings which opposes the flux of the machine's magnets and, as a result, the airgap flux can be reduced. This in turn reduces the back emf of the motor from equation **(1.3)**, making it possible to drive the machine beyond base speed, a technique known as field- or flux-weakening. Saliency ratios of $L_d < L_q$ generally improve field weakening performance due the reluctance torque present from increasing d-axis currents, and therefore are suited to higher speed operation [13], [17]. Conversely, non-salient machines or saliency ratios of $L_d > L_q$ have a comparably poor field weakening performance but can generally produce higher low speed torque. This can be because of a higher airgap flux density as the magnets are on the surface, not buried within the rotor structure, or a component of reluctance torque is generated in addition to the magnetic field attraction under q-axis current (negative saliency). To provide a summary of the findings in this chapter, Table 1 characterises the general machine parameters discussed for the Series-Parallel and Star-Delta winding configuration methods.

III. THE IMPLEMENTATION OF RECONFIGURABLE WINDINGS

Despite the early applications of star-delta and 'pole changing' induction motors [11], [18], [19], the main interest in drives with reconfigurable windings was focused on extending the speed range of permanent magnet (PM) machines the 1990s-2000s [10], [11], [20], [21]. This is linked with the shift away from asynchronous/induction motors in the automotive sector, due in part to an increased availability of rare earth magnetic materials, and the increased power

TABLE 1. Generalised influence of winding configuration on machine parameters.

Reconfiguration Method	Winding parts per phase	Stator leads	Switches	Connection	Flux (ϕ)	Inductance (L)	Resistance (R)
Star-Delta	1	6	5	Star	1	1	1
				Delta	$1/\sqrt{3}$	1/3	1/3
Series-Parallel	2	12	8	Series-Star	1	1	1
				Parallel-Star	1/2	1/4	1/4
Combined series-parallel star-delta	2	12	14	Series-Star	1	1	1
				Series-Delta	$1/\sqrt{3}$	1/3	1/3
				Parallel-Star	1/2	1/4	1/4
				Parallel-Delta	$1/2\sqrt{3}$	1/12	1/12

density and efficiencies the PM machines offered [22]. At this time, the predominant non-salient Surface Mounted Permanent Magnet (SMPM) rotor structures were generally not well suited for field weakening operation [20], [23]. Hence, efforts to extend the speed range of permanent magnet motors were divided into two strategies; One category of research endeavoured to improve the field weakening performance of PM machines through changing the structure of the rotor and therefore saliency of the machine, (usually with the addition of more complex control strategies) [17]. Other research entailed altering the voltage-speed relationship of the motor and inverter by either boosting the bus voltage [24], combining star delta windings [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35] or altering the winding structure to lower the back-EMF machine (as will be discussed in this review). If comparing further methods of electric machine speed extension methods is desired, the following sources are recommended [14], [36], [37].

A. RECONFIGURABLE WINDING METHODS

The most common reconfigurable winding methods covered in literature have been classified as star-delta, series-parallel, and tapped winding configurations. Suggested circuit diagrams are displayed in Figure 5, adapted in part from [37]. Each diagram shows how switching components can be arranged to achieve two different configurations of the motor windings when driven by a single inverter. For the series-parallel example: when all the switches marked as *series* are closed and switches marked *parallel* are open, the windings in each phase are connected in series. When all the parallel switches are closed and series switches opened, the windings in each phase are connected in parallel. This logic applies to the star-delta and tapped winding diagrams. Unless specifically desired as a feature, the switches for both configurations should not be allowed to be closed simultaneously during operation as this would 'short' the windings, bridging the phases together and in effect magnetically lock the rotor.

Another feature of either of the three circuit diagrams of Figure 5, is the ability to open any combination of the switches (if the user has the required level of control). As a result, the machine (or parts of the machine) can be isolated from the drive and implemented as a fault tolerance feature [38], [39], [40], [41], [42], [43], [44], [45], [46]. If a single phase of the motor is damaged or inoperable, it can be separately isolated from the drive by opening all c switches on that phase. This would allow the remaining two phases to continue to be driven, albeit with greatly reduced performance. If all the switches are opened, the motor can also be completely isolated from the drive, presenting a potential fail-safe feature for issues with either the motor or inverter. If the reader would like a review on the modification of machine topologies for fault tolerance, the following reference explores this in more detail [46].

Of the three methods shown, the most pertinent to this study is the series-parallel and star-delta methods as all the copper within the motor remains 'active' and carries current during both configurations. Tapped windings, despite having merits in terms of reducing motor losses [36], [37], can be explained as having a combined configuration: where the all the windings are used within the motor (alike the series configuration), and a tapped configuration: where half the windings are open circuit, thus reducing the number of turns per phase but not making a parallel connection which would further reduce resistances and inductances while also increasing the cross-sectional area of the conductors. Because of this, the series-parallel and star-delta methods present a higher potential for system performance enhancement and thus the focus later on.

The circuit diagrams of Figure 5 assume that the motor will be driven from a single inverter. If two inverters are considered for driving the machine, a different strategy for achieving multiple configurations can be employed. Star and delta connections can be made by connecting the other opposing ends of the phase windings to additional inverter as shown in Figure 6. Such a method is more commonly referred to as 'open-end windings' [36], [42], [43], [44], [45], [47], [48], [49], [50], [51]. Despite the open-end winding method having no specific reconfiguration equipment, it is evident that a second inverter and additional control strategies are required [36]. From a total of 8 semiconductor switches are necessary to achieve the series and parallel connection while the star and delta configurations are possible with only 5 switches. Two inverters have also been used to

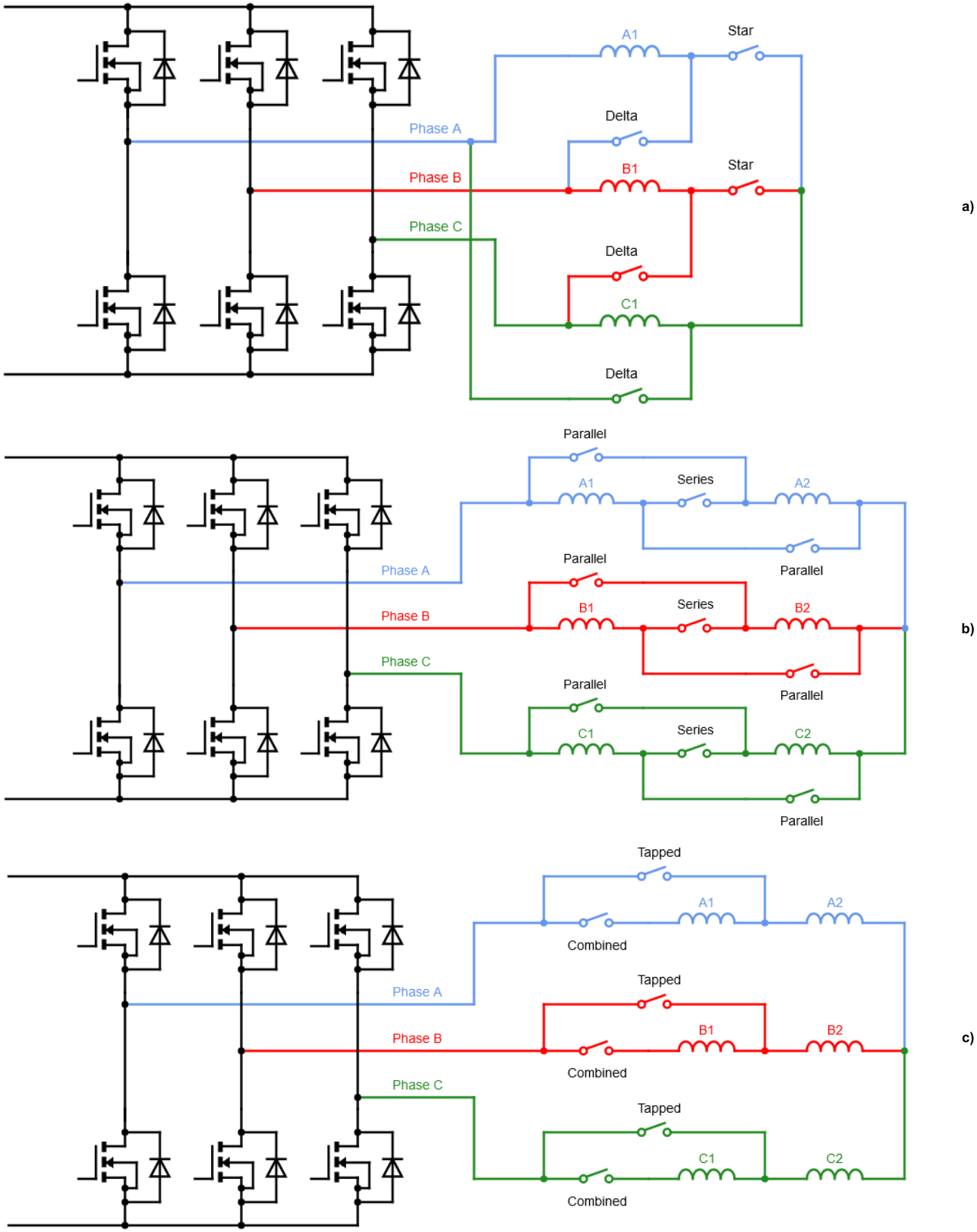


FIGURE 5. Circuit diagram for the reconfigurable equipment, a) Star-Delta, b) Series-Parallel, and c) Tapped Windings when driven from a single inverter.

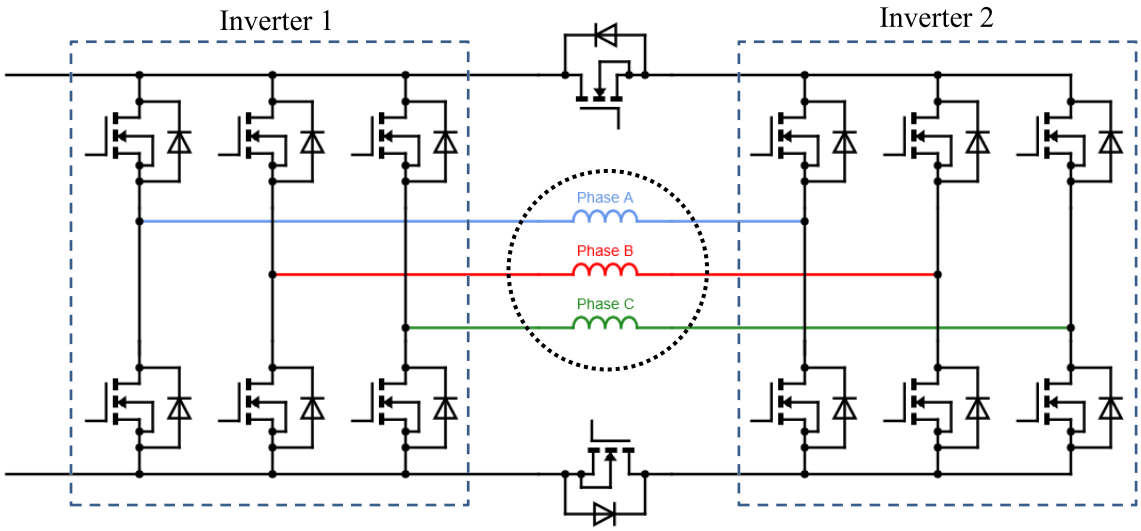


FIGURE 6. The 'open-end windings' strategy for star-delta reconfiguration using two inverters.

achieve series and parallel configurations, the most recent development of which is presented by Tang L et al of Oakridge laboratories, Figure 7 [52].

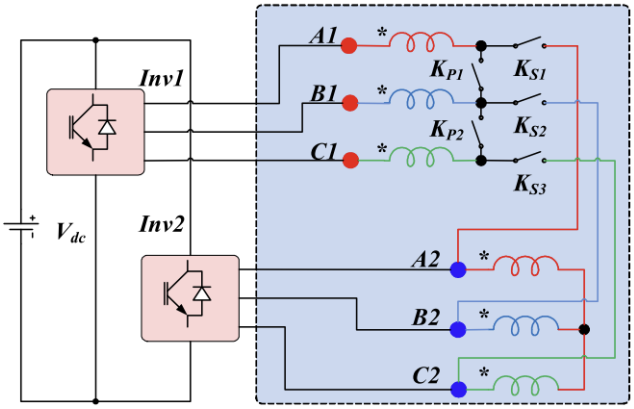


FIGURE 7. Dual inverter strategy to achieve series-parallel winding configurations using 5 additional semiconductor switches [53].

The power components of each inverter are designed to have half the rated current compared to a single inverter driven motor, partially mitigating the use of multiple inverters. The switching frequency can also be doubled by obtaining the optimal phase shift, the result of which significant reductions in torque ripple can be made [54]. Such a design still requires 5 semiconductor switching components to operate in addition to the second inverter (KP1, 2 for the parallel connection and KS1, 2, 3 for the series configuration). This of course increases system costs and complexity as well as representing another source of parasitic losses in the additional inverter and semiconductor switches used.

B. STAR/DELTA VS SERIES/PARALLEL

Early investigations into the implementation of reconfigurable windings for traction drive applications [13], [20], [55], show that the ratio of torque and speed between series-parallel configurations is proportional to the ratio of flux generation; the series connection achieving approximately twice the torque, but half the base speed of the parallel connection. This ratio is greater than the star-delta method; where the star configuration has approximately 1.73 times the torque capability than the delta configuration. This demonstrates that the series-parallel method has a greater capability for speed and torque extension. When considering the complexity of reconfiguration equipment for both methods however, the star-delta method requires fewer stator leads and switching components, lowering the overall size and packaging requirements [56]. For the star-delta method, access to each end of the phase windings is required to achieve both configurations, the advantage being that these are usually accessible on a conventional machine and can be modified without removing the existing winding structure. However, if increasing the flux density above the baseline value is required this will necessitate rewinding the motor to include a higher number of turns than before.

To summarise the generalised traits of the three winding configurations discussed, Table 2 is populated with a comparative value based on the findings of this review.

TABLE 2. Comparative traits of reconfigurable winding techniques.

	Complexity	Performance	Cost	feasibility
Star delta	**	***	**	*****
Series Parallel	****	****	****	***
Tapped Windings	***	**	**	***

IV. APPROACH 1: REDUCING E-DRIVE SYSTEM MASS AND VOLUME

This section will only cover the recent development of 'reconfigurable winding' technology with the ability to change configuration during operation. In its simplest form, the volume of an electric machine is proportional to its power capability [15], [57]. This means that when the power requirement is fixed by the application, the main consideration of the winding topology is obtaining the desired torque and speed range. Requirements of traction drives are characterised by the ability to provide a very high transient torque, but also cover a large speed range [58]. In applications where the final drive ratio can be changed, this can be used to change the torque speed relationship but there will still be a trade-off between the torque and speed capability for a fixed winding structure. Multispeed mechanical gearboxes are a possible solution to this but introduces additional weight, losses, and cost [55], [59]. The reconfigurable winding method offers an electromagnetic alternative to a multispeed gearbox, without the additional mechanical equipment. One of the mutually agreed upon aims for implementing reconfigurable windings covered in literature is reducing the mass and volume of the drive system (**TABLE 3**). A common methodology, is to maintain the torque and power rating the motor whilst reducing its volume and size [53], [60], [61], [62], [63]. As changing the winding configuration has shown to alter torque density, the volume of the motor can therefore be reduced whilst producing a similar performance. Other than redesigning, or choosing a smaller machine, one accessible method of achieving this is through shortening the axial length of the machine [57], [60]. This action alone would naturally result in reduced torque output, however, an increased turns per phase indicative of the 'low speed' winding configuration reclaims this lost torque when operated with the same phase current as before, albeit over a shorter speed range. It has also been shown that the lower turns per phase of the 'high speed' winding configuration not only maintains the power output of the baseline motor but can exceed it in certain circumstances. To quantify the degree to which the motor volume can be reduced, a study using finite element analysis based on the Toyota Prius PMSM reported a 24% reduction in axial length while maintaining peak torque and power outputs [60]. Another study shows a volume reduction of 25% possible through the reduction in stator diameter while increasing the peak torque values by 15% [58]. Furthermore, theoretical limits for multiphase machines indicate that a 27% reduction in motor volume is feasible compared to that of a fixed configuration [57].

A. TORQUE DENSITY IMPROVEMENTS AND SPEED RANGE EXTENSION

When implementing multiple winding configurations to an existing design, which is most applications shown in TABLE 3, the ratio of winding turns compared to the baseline makes a large difference to the outcome performance of the drive. Using the series-parallel method as an example, there are three methods to choosing the new winding structure: If the series configuration has the same number of turns as the baseline (and thus the same characteristics), the parallel connection will primarily extend the speed and power capability of the motor. If the parallel configuration has the same number of turns as the baseline configuration, the series configuration will primarily increase the torque capability of the motor. The third method is to design the series configuration to have a higher number of turns, and the parallel configuration to have fewer turns than the baseline winding. This method of choosing a number of turns either side of the baseline configuration, increases the potential torque density, speed range, and power density of the motor. Because of this, it is the most popular method for demonstrating the advantages of reconfigurable windings in literature.

This ability to alter the torque density of the motor also influences other components in the drive system, of which the possible elimination of mechanical/hydraulic transmissions and their peripheral components show the greatest potential not only in terms weight reduction, but volume, loss reduction, and efficiency increase of the drive unit as a whole [11], [53], [55]. Mimicking the torque speed characteristics of a two-speed gearbox, a motor with reconfigurable windings lends itself to improving the viability of in wheel hub motors. Given their radial dimensional constraints, to provide the large tractive torque for low-speed acceleration, motors used in the 'hub' of a conventional passenger vehicle are required to be bulkier than geared their counterparts thus further research in this area targeted to ''achieve direct drive traction motors with extremely high torque capacities and low weights and volumes'' [61].

Low power automotive drives that require extended torque speed range characteristics have also been a key area of research of this technology [56], [63], [64], [66], most notably the development of a hybrid electric scooter that utilises reconfigurable windings to enhance the operation of a small motor generator [56]. Research on minimising costs in the drive system focus on limiting or completely excluding the operation of the motor in field weakening with the aim to reduce materials in the inverter cables/connections and permanent magnets in the motor. If the ability to use field weakening is maintained, changing the winding configuration will alter the operating characteristics of the motor in this region, and as machine design is heavily influenced by its operation in field weakening (or lack thereof) the topology of the motor itself has scope to be revised. This therefore offers further capacity for volume and mass reduction of not only the motor itself, but the proportion of materials used within them such as the permanent magnet mass and stator slot size [13], [62] [65]. Furthermore, the influence of changing the winding configuration may also have potential on reducing the size of the cooling system for the motor and inverter as they may be operating with different current densities to before [36].

The literature covered strongly suggests that the implementation of reconfigurable windings has the capability to

TABLE 3. Approach 1: Reducing E-drive system mass and volume.

Ref.	Aim	Winding	Motor	Switchgear	Application	% Increase in	
						Power Density	Torque Density
[52]	Increasing power density and reduce permanent magnet mass	SP	PMSM	Thyristor + Dual Inverter	Automotive	100%	41%
[37]	Comparisons of three reconfiguration techniques.	SP, SD, TW	PMSM	-	Automotive	50 -90%	-
[58]	Pole changing influence on Simulated torque/power density	PC	IM	Power electronics	Automotive	50%	47%
[64]	Field weakening control and armature current reduction.	SP, SD	BLDC	Contactors	Automotive	43%	53%
[36]	Impacts of reconfigurable windings on drive performance	TW	PMSM	SCRs, RB-IGBTs, IGBT-Diode	Automotive	84%	-
[65]	Increasing maximum torque and usable torque-speed	SP	PMSM	-	Locomotive	-	82%
[55]	Overload starting torque without increased inverter current rating.	SP	IM	-	Automotive	-	80%
[60]	Axial reduction of motor size maintaining power and efficiency.	SP	PMSM	Contactors	Automotive	36%	24%
[18]	Reducing core losses and speed range extension	PC	PMSM	Variable strength magnets	Automotive	-	59%
[66]	Extending the speed range of a BLDC motor.	TW	BLDC	Full/Half Bridge	Automotive	-	50%
[63]	Extending the speed range of an induction motor.	SP	IM	Solid state switches	Industrial	-	40%
[61]	Optimisation of drive volume for best torque-speed performance.	SP	DC	IGBT/MOSFET	Locomotive	-	38%
[57]	Volume reduction maintaining torque and power output.	SP, SD	PMSM	IGBTs/ MOSFETs	Automotive	27%	-
[67]	Increasing asynchronous torque, decreasing braking torque	Line Start	PMSM	-	Industrial	-	23.5%
[56]	Speed extension avoiding field weakening	SD	BLDC	Contactors*	Automotive	20%	-
[68]	Reducing harmonics and NVH	Combined SD	PMSM	no switching	Automotive	-	9%
[20]	Drive speed range extension whilst avoiding field weakening	SP, SD	SM-PMSM	Thyristor	Automotive	-	-
[69]	Alternative to field weakening to extend the speed range	SP, SD	PMSM	Contactor/ Thyristors	Locomotive	-	-
[13]	Drive speed range extension whilst avoiding field weakening	SP	PMSM	Contactor and thyristor	Locomotive	-	-
[70]	Calculation of torque-speed-characteristics for coil switching	SP, SD	PMSM	IGBT / MOSFET	Automotive	-	-
[10]	Constant power range and switching time improvements.	TW	IM	IGBTs	Machine tooling	-	-
[62]	Doubling the field weakening range of a PMSM	DI	PMSM	Thyristors	Automotive	-	-
[71]	Calculations for starting currents and torques	SD	Induction	Contactors	Industrial	-	-

__PMSM__: Permanent Magnet Synchronous Motor, __SMPMSM__: Surface Mounted Permanent Magnet Synchronous Motor, __BLDC__: Brushless Direct-Current , __DC__: Direct Current, __IM__: Induction Motor, __SRM__ Switched Reluctance Motor. __SP:__ series-parallel, __SD__: Star-Delta, __TP__: Tapped Windings, __DI__: Dual Inverter __PC__: Pole Changing

Where power densities (kW/kg) are not directly disclosed or compared with that of a fixed winding case, these values are derived from other metrics or omitted entirely.

bring system efficiency gains due to lower losses in motor and inverter over specific operating conditions (which will be discussed later). Relatively small efficiency gains have large implications on vehicle range and energy storage considerations. A large proportion of the mass of a typical BEV is the battery itself, furthermore the mass of the vehicle has been shown to be linearly proportional to energy consumption [72]. The decrease in energy consumption due improved drive efficiencies could allow the downsizing of the battery system while maintaining the baseline vehicle range. Depending on the level of mass removed from the battery, this could further decrease the energy consumption of the vehicle over the drive cycle, allowing for an iterative reduction of vehicle mass until a new optimum is achieved.

An equivalent motor and reconfiguration equipment has been shown to fit in the baseline motor housing at a reduced mass [60], [73]. In hub motor applications this has the benefit of reducing the unsprung mass of the wheel assembly improving vehicle platform dynamics [61]. Despite these studies, adding the necessary switchgear required to reconfigure the windings of the stator switchgear will incur the additional mass and volume of the required components. Methodologies for achieving the switching capability varies greatly with one of the major challenges being the placement of the switchgear and increased phase leads to be terminated outside of the motor casing [56]. Conversely, one study has shown that all the peripheral components required to achieve winding reconfiguration along with the inverter can be integrated within the motor itself [73].

B. INCREASING MOTOR POWER DENSITIES

If the power output of a traction drive is limited by the current density (A/mm^2) of the motor windings, using configurations with a lower number of turns than the baseline winding structure allows for a significant increase in the high-speed power density of the motor [36], [37], [52], [58], [64]. This is because the cross-sectional area (mm^2) will increase when winding parts are connected in parallel (2 times) or delta (1.73 times), thus the current(A) delivered to the motor can increase by the same amount keeping the current density of the phase windings constant. Simulations of [52] and shown in Table 1, demonstrates that the series-parallel method using half the half the number of turns of the baseline motor in the parallel configuration can improve high-speed motor power densities by 100%. Although the potential power output of the motor increases, to achieve this inverter must be able to deliver a higher phase current than before. Moreover, the battery system will also be placed under a higher load, thus the realistic implementation to increase the power of existing drives would require higher rated components. Conversely, this demonstrates the possibility for drastically reducing motor sizes while maintaining the baseline output powers.

In cases where output power is limited by inverter current, a configuration with a higher number of turns could improve the low-speed power capability of the motor without increasing the line/inverter current or any other modifications to the drivetrain [55]. This can be used to better match the power capability of the motor to the power capability of the inverter, changing to a parallel or delta configuration to maintain that power level at higher speeds [60]. The phase current density in the motor does however increase in line with the phase resistance which may push the existing saturation or thermal limits of the motor design. Because of this, the increase in power output may not be a linear correlation and the motor would be expected to reach its thermal limit faster.

V. APPROACH 2: IMPROVING E-DRIVE EFFICIENCY

Reconfigurable windings show promise of improving the efficiency of the electric drive. Although multiple authors have researched the generalised influence [13], [57], [70], the exact benefits to be gained from changing the winding configuration is shown to vary between motor topologies and applications leading to a vast difference in aims and results in literature shown in TABLE 4. This section will reflect on the objectives and methods applied to improve drive efficiencies through the implementation of reconfigurable windings for traction drive applications. One of the key objectives for improving the drive system performance is to reduce losses in the drivetrain, with a particular attention to the transmission [12], [53], [56], [64]. The ability to alter the torque speed relationship of the motor through winding reconfiguration achieves a comparable characteristic of a multi-speed gearbox and if used in its stead, can have a significant impact on driveline efficiency. The exclusion of frictional losses associated with the continuously variable transmission (CVT) is one application with the largest impact on improving driveline efficiencies without the need of any other speed extension method such as field weakening (incurring additional cost and complexity to the controller) [56], [62], [64].

From TABLE 4, the implementation of reconfigurable windings has largely been applied to reducing losses in the motor windings. Copper losses are reduced in part due to the reduced phase resistance [36], [37], core losses are also reduced due to the smaller field coupling of the high-speed winding [61]. The greatest reduction in motor losses is seen at high-speed operation largely attributed to the reduction in required d-axis current to extend the speed range of the motor, i.e avoiding field weakening operation [20], [56], [64]. The result of this culminates in the greatest loss reduction compared with other common methods of speed extension as shown in Figure 8 [37]. Approximately 50% motor loss reduction is common in literature and is found at high speed and high load operating conditions, showing potential for improving the thermal capabilities and lowering cooling requirements [8], [18], [36], [37], [56].

Although only covered in a few cases [9], [53], reconfigurable windings also show promise for significantly reducing inverter losses in low and high-speed configurations. When considering the efficiency of the whole drive system, these losses can make a significant contribution in the reduction

TABLE 4. Approach 2: Improving E-drive efficiency and loss reduction.

Ref.	Aim	Winding	Motor	Switchgear	Application	% Change in	
						+ Efficiency	- Loss
[61]	Optimisation of drive volume for best torque-speed performance.	SP	DC	IGBT/MOSFET	Locomotive	-	75%
[56]	Speed extension avoiding field weakening and associated control complexity for low power cost-sensitive applications.	SD	BLDC	Contactors*	Automotive	3%	61%
[37]	Comparisons of torque, output power and losses of three reconfiguration techniques.	SP, SD, TW	PMSM	ND	Automotive	8-10%	50%
[8]	Increase drive efficiency and reduce the phase current ripple for a given switching frequency.	-	Polyphase BLDC		Marine	-	59%
[36]	The test results show that RW system can achieve up to 84% power boost with 57% copper loss reduction in high speed	TW	PMSM	SCRs, RB-IGBTs, IGBT-Diode	Automotive	-	57%
[18]	Reducing core loss by 49% or 33% and tripling the speed range of the motor	PC	PMSM	Variable strength magnets	Automotive	-	33-49%
[52]	Increase motor power density and reduce permanent magnet mass	SP	PMSM	Thyristor + Dual Inverter	Automotive	10%	28%
[53]	A reduction of motor size for a given power rating and comparisons with mechanical transmissions.	SP	PMSM	Thyristor + Dual Inverter	Automotive	1%	35%
[64]	Cost reduction of BLDC: Elimination of field weakening control and reduction in armature current.	SD	BLDC	Contactors	Automotive	31%	-
[60]	Axial reduction of motor size maintaining power and efficiency.	SP	PMSM	Contactors*	Automotive	1%	9%
[74]	Winding configuration switching algorithm to achieve maximum drive efficiency	Open End Winding	PMSM	Dual Inverter, Dual Power Source	Automotive	3%	6%
[75]	Wide range speed control of permanent magnet (PM) brushless motor drive	Phase-decoupling	Polyphase BLDC	ND	Automotive and industrial	-	-
[76]	Avoidance of torque variation when reconfiguring the winding from series to parallel and from star to delta	SP, SD	BLDC	Relay	Industrial	-	-
[12]	Elimination of mechanical transmission with a reconfigurable switched reluctance motor.	SP	SRM	Relay	Domestic	-	-
[51]	take into consideration the power segmentation, redundancy degrees in degraded mode, the THD current and the torque undulation	Open End Winding	IM	Dual Inverter	Industrial	-	-

***PMSM**: Permanent Magnet Synchronous Motor, **SMPMSM**: Surface Mounted Permanent Magnet Synchronous Motor, **BLDC**: Brushless Direct-Current , **DC**: Direct Current, **IM**: Induction Motor, **SRM** Switched Reluctance Motor. **SP:** series-parallel, **SD**: Star-Delta, **TP**: Tapped Windings, **DI**: Dual Inverter **PC**: Pole Changing*

of energy consumption [9], with total drive system efficiency improvements reported of approximately 8-10% [37]. In series or star configurations with a higher number of turns than the baseline, a lower phase current is required to generate

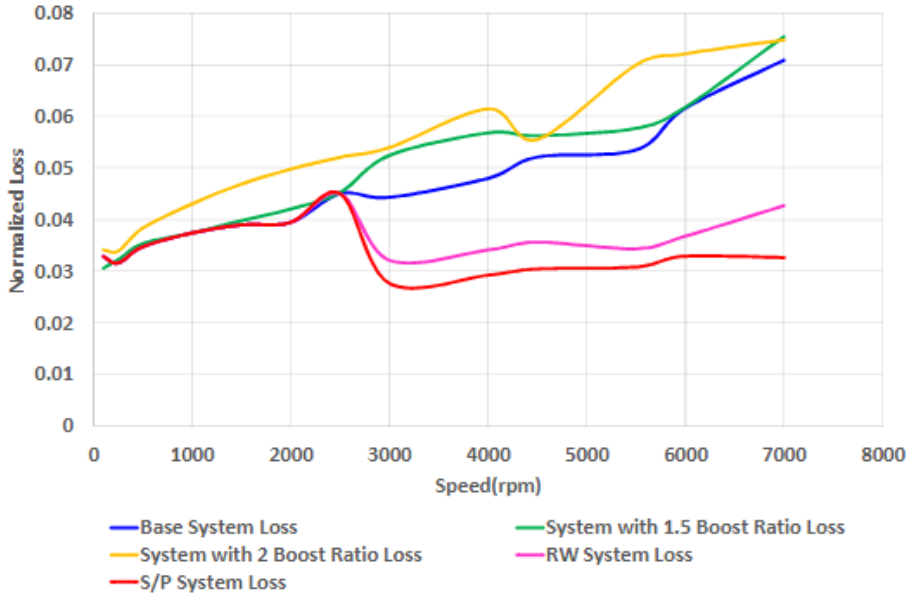


FIGURE 8. Efficiency comparison of speed extension methods [37].

the same torque below base speed. This reduction in phase current translates to lower losses in the inverter(s). As the semiconductor power switches have a characteristic resistance, ohmic losses of I^2R show that a modest reduction in current will have a large influence on semiconductor heating and associated losses. The result of this is a clear increase in system efficiency at low-speed operation as demonstrated in Figure 9 Influence of changing winding configuration on the total system efficiency.

In high-speed configurations, due to the reduced back emf, the motor is operating with less or no field weakening compared to the low-speed configuration and the d axis current component is small or not necessary at all. During this condition the line current will again be smaller than before meaning that the associated inverter losses are reduced [36]. This is reflected as a secondary high system efficiency zone in Figure 9 and a slight increase its maximum efficiency [60]. This shows potential for reducing the cooling requirement for the power electronics at both high and low speed operation.

The important implication for automotive applications is that when applied to an existing drive, both star-delta and series-parallel methods offers a higher efficiency region for both low and high-speed operation. The series and star configurations could be implemented to reduce 'city' driving energy consumption, while parallel and delta configurations also increase high-speed drive cycle efficiencies in line with motorway driving [9]. Improvements in total system efficiencies under motorway driving conditions could have a large impact on increasing vehicle range or allowing the downsizing of powertrain components.

A. ADDITIONAL LOSSES

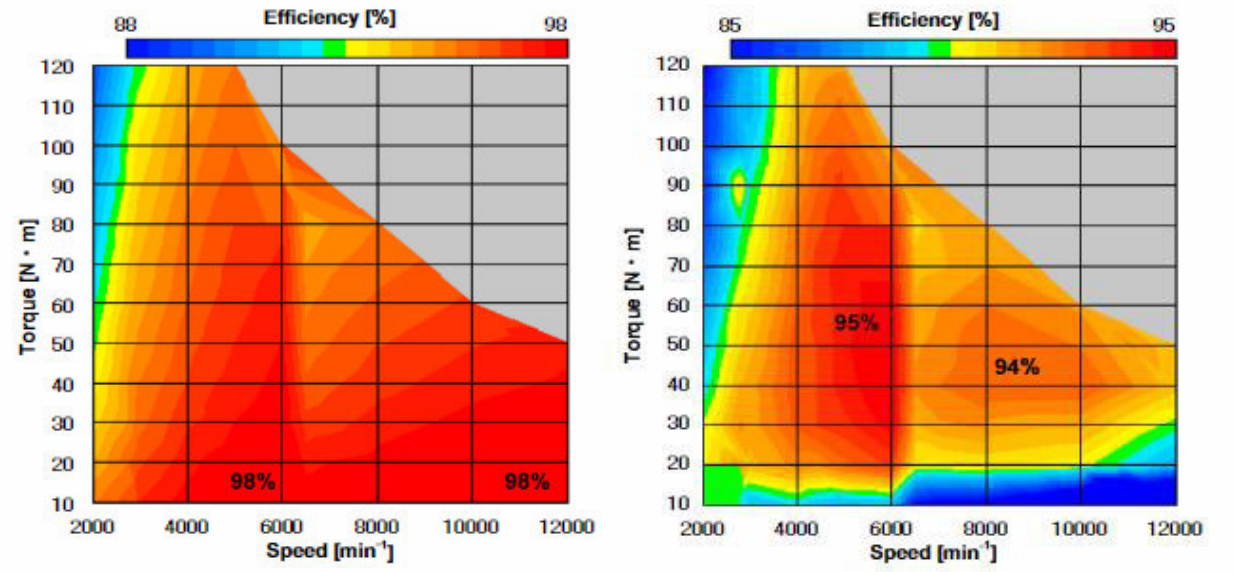
For the star-delta, series-parallel and tapped winding methods, current will be carried across parallel phase windings. Depending on the placement of these parallel windings and imperfections in the winding of the motor, one may see a higher flux than the other. In this circumstance, due difference in emf generated by the coils, a current will be induced between them when a voltage is applied across the phase terminals from the driver generating unwanted losses in the windings [77]. Admittedly, these losses are usually negligible although research undertaken quantifying the effect of circulating currents in parallel windings state that they can become significant in high-speed machines. Luckily it has been shown that by twisting the parallel windings together the difference in phase amplitude in the windings is drastically reduced and with it the circulating currents in them [77]. Delta configurations reportedly suffer the most from third harmonic currents with the adoption of non-salient rotors essential to mitigate this issue [69].

As phase current will be carried through any reconfiguration equipment, any additional resistances due to the switching components or cabling will be a source of ohmic losses. If semiconductors are used as the switching components, the associated losses follow the general equation of:

$$P = V_{TO}I + rI^2$$

where $\boldsymbol{P}$ is the power losses, $\boldsymbol{V_{TO}}$ is the semiconductor turn on voltage, $\boldsymbol{I}$ is the phase current and $\boldsymbol{r}$ is the on-state resistance. [20].

The advantages of using semiconductor switches is the ease of implementation, lack of maintenance, and wide control flexibility [14]. Mechanical switching devices on the other hand can be designed to have practically negligible losses [20], [69], have high temperature tolerances and can be easily manufactured but also have a longer associated switching transient where motor torque will be interrupted.



a) Combined Inverter and configuration equipment efficiency

b) Total drive efficiency

FIGURE 9. Influence of changing winding configuration on the total system efficiency [9].

B. CIRCULATING CURRENTS

The star-delta, series-parallel winding configurations both require the phase current to be carried across two windings when configured in the 'delta' or 'parallel' configuration. It is stated that due to the placement of these parallel windings and imperfections in the winding of the motor, one coil may see a higher flux density than the other. In this circumstance due difference in emf generated by the coils, a current will be induced between them when a voltage is applied across the phase terminals from the driver generating unwanted losses in the windings [77]. Admittedly, these losses are usually negligible although research undertaken quanifying the effect of circulating currents in parallel windings state that they can become significant in high speed machines. Luckily it has been shown that by twisting the parallel windings together the difference in phase amplitude in the windings is drastically reduced and with it the circulating currents in them [77].

VI. FURTHER CONSIDERATIONS

Although the implementation of reconfigurable windings show promise an improved high-speed performance and efficiency for the machine, there are a few considerations that do not directly fall within scope of improving torque and power density or loss reduction.

A. COST REDUCTION

RW machines with the capacity to generate higher flux densities result in a lower rated current for the same torque output [36]. A reduction in phase current allows for downsizing of semiconductor materials [73], cables motor windings, and other 'phase current' rated components resulting in cost savings in addition to overall operational costs from lower energy consumption [36]. There is also speculation as to whether this saving in inverter semiconductor materials could offset the additional cost of the RW switchgear [58]. Furthermore, as the motor and inverter can be operated at lower currents, a reduction in heat can be expected further contributing to lowering system costs if the cooling system can be simplified [36].

B. PERMANENT MAGNET REDUCTION

Methods of implementing reconfigurable windings to increase the torque density of electric motors can also be used to reduce the mass of permanent magnet materials in the rotor while maintaining baseline performance [52]. Moreover, this technique has been shown to improve the performance of induction and switched reluctance motors closer to the performance of a PMSM [12]. This shows promise of reducing one of the largest costs in the production of electric machines but also the a reduction in life cycle emissions and material depletion related to these rare earth metals.

C. TORQUE INTERRUPTION

With the severity of delay depending on the switching architecture employed, torque interruption when changing from one configuration to another is a common trait inherent of the winding reconfiguration method and an area of further research [51], [76]. All methods covered in the literature require all phase currents to be zero during the switching event to avoid large destructive voltage spikes in the switching assembly [53], [64]. The focus in literature discussing this torque interruption is to reduce the switching event as much as possible to minimise the discomfort of the vehicle passengers [20], [69]. An early example of this

switching transient for a mechanical contactor is to be between 55-75ms. 4-10ms is the measured equivalent for a semiconductor switching array [69].

VII. APPLYING RECONFIGURABLE WINDINGS TO REDUCE INVERTER CURRENT: A CASE STUDY

The 2015 Nissan Leaf represents a well-established electric vehicle on the market today along with a well-developed propulsion system, the architecture of which can be generalised to most existing passenger vehicles. It is because of this, and the freely available information on the motor and inverter, that is has been chosen to showcase a potential application of reconfigurable windings to an existing architecture, following a mixture of approaches 1 and 2 covered in this review. In electric vehicles, the inverter plays a vital role in converting DC power from the battery into a suitable waveform to drive the electric motor. The sizing and specifications of the inverter are required to meet the torque and power requirements of the motor(s). It is observed that for the powertrain of the Nissan Leaf, the combination of motor and drive necessitates a high peak inverter current rating compared with the peak power demand of the vehicle. This apparent over sizing of the inverter is to satisfy the peak torque rating of the motor, as the motor demands a significantly elevated current from the inverter in this operating range.

The following section will briefly discuss the application of series-parallel winding configurations to the Nissan leaf traction motor to reduce the current demand from the inverter: The proposed design maintains the baseline number of turns to act as the parallel configuration, targeting low load and high-speed operation. The series configuration will therefore double the effective turns per phase increasing the ampere-turns by a factor of 2. Modelling and subsequent simulations were carried out in ANSYS MotorCad, In which the magnetic structure of the stator and rotor remain unaltered between the baseline and proposed reconfigurable winding design so that the only variable in question is the number of turns per phase. Two independent models were developed to represent the 'series' and 'parallel' configurations and were solved separately. Therefore results shown later are a combination of two discrete winding configurations favouring the parallel configurations for torque demands below 150Nm. The main specifications used for the baseline Nissan leaf drive are shown below in TABLE 5.

TABLE 5. Baseline 2015 NISSAN leaf motor key parameters.

Bus Voltage	V	375
Peak Inverter Current	Arms	480
Turns per Phase	n	24
Phase Resistance	Ω	0.01106
Phase Current Density	A/mm^2	16.88
Line Inductance	mH	0.409
Peak Torque	Nm	287
Peak Power	kW	80
Maximum Shaft Speed	rpm	10,000

It should also be stressed that there are additional constraints regarding the control of a machine with reconfigurable windings. An example of which being that it is desirable for the reconfiguration event to happen within the base speed of the mode with the highest BEMF constant. Switching above base speed has not been demonstrated in literature and is problematic because the control of Id currents is not possible during the changeover transient. And therefore it is likely to incur high voltage transient events which may damage switching components. it is not the intension to discuss the practical implementation further however there is examples of research on this matter is available to the reader [78].

Using the peak current capability of the inverter and operating voltage of the DC bus, the inverter can be estimated to have a peak power capability of 180kW. However, the vehicle is limited to 80kW output indicating that the inverter is oversized which is common place in automotive applications. This is due to the high transient torque requirement for acceleration and gradability but a comparatively low power requirement at higher speeds. This is demonstrated Figure 10 showing **a)** the torque speed efficiency curve of the baseline motor and **b)** subsequent current demand with a very high current region indicated for operation above 150Nm. This high current region could hypothetically be offset through the modification of the final drive ratio, a change in the number of turns of the armature winding, or the addition of a mechanical gearbox. However, each of these methods have negative implications such as higher mechanical stresses, lower machine efficiencies, and the requirement of additional mechanical components and respective mass and complexity.

The proposed implementation of series and parallel configurations (TABLE 6) provides a further alternative for this scenario, reducing the inverter current requirement without the inherent drawbacks of the aforementioned methods. The higher torque constant achieved by the series configuration provides the same peak torque value as the baseline machine, with a reduction in inverter current of 41.67%. The torque curve in the series configuration is characterised by a slight reduction in peak power output due to the shorter base speed region and lower efficiency operation beyond this point. The parallel configuration performance is identical to the baseline as it is has the same number of turns.

Despite the peak inverter current reducing from 480-280A, as the phase resistance doubles thus maintaining the same phase current density as before. It is therefore assumed that the motor can safely operate with the winding modifications both thermally and mechanically as there are no modifications or required change in the rotational speed of the rotor.

The operating strategy currently displayed depicts the parallel configuration as the default state under a torque demand of 150Nm as it provides the highest motor efficiency. However, depending on the potential inverter loss reduction in the series configuration, the net losses between the motor and inverter may point to the series configuration as the optimum state for low speed efficiency. The caveat to this remains the losses associated with the reconfiguration equipment yet to be

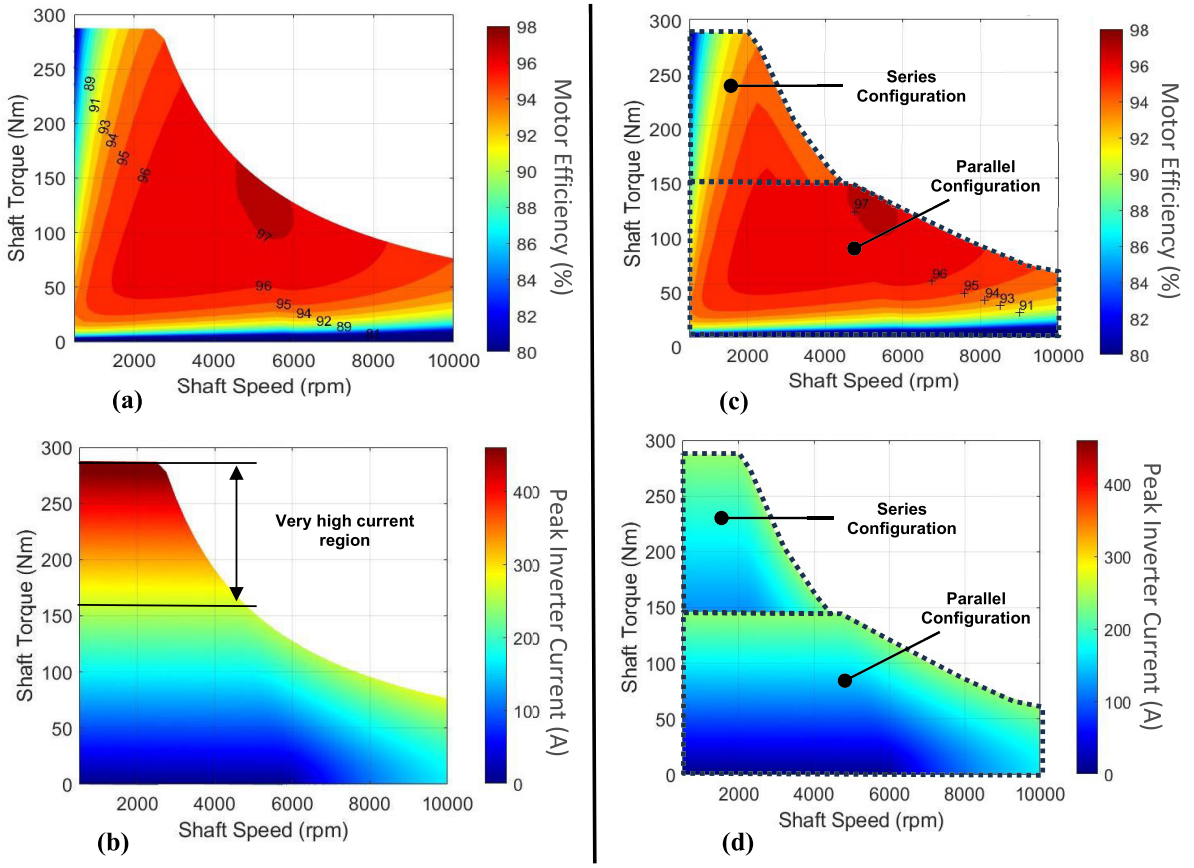


FIGURE 10. Baseline (left) 2015 Nissan leaf electric machine with (a) representing the torque speed efficiency map and (b) showing the corresponding peak inverter current demand. The machine with reconfigurable windings (right) with (d) showing significant reduction in phase current versus the region highlighted in (b).

fully scrutinised. There are further system level implications regarding the control strategy of the winding configuration state; the improved control over inverter thermal management gained by the potential phase current reduction, advantages gained in motor control due to higher inductances, and additional fault tolerance mechanisms.

The preliminary results show that reconfigurable windings can reduce the rated current required of the inverter of the Nissan Leaf E-Drive by 41.67%, insinuating possible size, weight, and cost reductions can be made while maintaining the same level of performance. This case study highlights the potential of reconfigurable windings as a promising technology for the development of more efficient and cost-effective electric vehicles. Furthermore, the influence on inverter losses and thermal management are suspected to be significant when compared to current inverter implementations. This outlines one possible methodology to the application of reconfigurable windings previously discussed in literature, however the further considerations regarding the integration of this technology are still to be explored.

VIII. CHALLENGES AND KNOWLEDGE GAPS

This review paper has outlined the theoretical aspects and potential performance benefits of traction motors with reconfigurable windings. To implement this technology in a practical application further research is required in two key areas: first, control strategies for motors with reconfigurable windings, and second, potential advantages and disadvantageous regarding the functional safety of the switching components. Control methodologies should be explored to reduce total BEV energy consumption over driving scenarios and be leveraged to improve vehicle performance. The control strategy and switching transient will

TABLE 6. 2015 NISSAN leaf motor using series-parallel reconfigurable windings.

Bus Voltage		V	375
Peak Inverter Current		A	280
Turns per Phase	Series	n	48
	Parallel	n	24
Phase Resistance	Series	Ω	0.04424
	Parallel	Ω	0.01106
Phase Current Density	Series	A/mm^2	16.88
	Parallel	A/mm^2	8.44
Line Inductance	Series	mH	1.636
	Parallel	mH	0.5892
Peak Torque		Nm	287
Peak Power		kW	75
Maximum Shaft Speed		rpm	10,000

also impact the driver's comfort and driving experience. The hardware in place to enable switching from one configuration to another will likely also impact functional safety, either through introducing increased complexity, leading to higher likelihood of failure, or by offering the potential to mitigate motor failures by implementing fail-safe configurations.

Further research on total system losses would help draw more conclusions on real world efficiency gains to be gained from traction motors with reconfigurable windings. This is particularly true of the losses associated with the switching components. Comparing mechanical switching components to semiconductor equivalents is crucial for forecasting the cost-effectiveness of this technology, as the cost of semiconductor power devices, or the requirement of multiple inverters, will most likely hinder implementation. Future studies could also focus on applications with a view to reduce motor and drive cost. As the literature has shown that multiple winding configurations can increase torque and power densities (primarily of PMSMs), this could allow for a reduction in motor volume or permanent magnet requirement. Furthermore, this could enable less favourable motor topologies in the automotive sector (SRMs for example) to be a more attractive proposition.

The majority of literature discuss the modification existing machines. if we consider the initial winding configuration can now be modified it is almost cirtain there will be a new optimimum. Furthermore, there will be other variables during the design process that will be influenced by the flexibility reconfigurable windings offer to the designer if intoduced propectively. It is therefore recommended that further research be focus to the methodology of desiging machines with reconfigurable windings from conception to provide insight on true applicability.

IX. CONCLUSION

A full review of reconfigurable windings for traction drives has identified three key areas for consideration when evaluating their benefits: their influence on reducing e-drive system mass and volume, their role in improving e-drive efficiency, and methodologies for implementation.

Studies that directly discussed improvements in motor performance commonly found machine torque and power density improvements of 40-80% when implementing reconfigurable windings to existing machine designs. Other studies targeting efficiency achieved total motor loss reductions of 30-60%. A new case study on the Nissan Leaf drive highlights yet other potential impacts on the drive system as a whole, achieving a reduction in inverter current requirement of 41.67% while maintaining baseline performance characteristics.

Studies that discuss the reduction in motor losses very rarely consider system impacts, such as the influence on the inverter or losses incurred within the switching architecture. For example, many studies have used semiconductor-based implementations to reduce the transient switching (reconfiguration) time without considering the added on-state resistance, which may negate overall system efficiency benefits in some applications.

Mechanical switching components offer an attractive solution for automotive traction applications due to their simplicity and low material cost compared to semiconductors. However they are rarely explored in the literature due to their higher transient switching time. Future work considering mechanical switches for reconfigurable windings must ensure that the contact resistance of the switch is low enough as to retain total system efficiency benefits.

Overall, the general applicability of reconfigurable windings is unclear due to the reliance of previous literature on specific case studies and retrofitted machines. There is a need for an overarching methodology for the implementation of this technology in the early stages of machine design, and structured analysis to identify which applications may benefit the most.

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